

Securing the *Edge*: An AI-Driven Detection of Intrinsic Cyber Threats via Current-Profiling in IoT Networks

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Abstract— While Internet of Things (IoT) devices are traditionally perceived as passive network components performing routine tasks with minimal network interaction, this assumption creates a dangerous security blind spot. Malicious actors can exploit manufacturing processes or third-party vendor channels to embed hardware Trojans that remain dormant until triggered to execute programmed attacks. These Hardware Intrinsic Attacks (HIAs) represent an emerging and particularly insidious threat vector that has garnered significant attention in the cybersecurity community. This paper presents an experimental intrusion detection system (IDS) specifically engineered for IoT and Edge computing environments, addressing critical security gaps through hardware-based monitoring approaches. The study evaluates four distinct cyberattack categories: Covert Channel Attacks (CCAs), Power Depletion Attacks (PDAs), Denial-of-Service Attacks (DoS), and Man-in-the-Middle Attacks (MIMAs), systematically implemented across a heterogeneous testbed comprising ESP32 microcontrollers as IoT sensor nodes, Raspberry Pi 5 systems, and Nexys2 Field-Programmable Gate Arrays (FPGAs) as Edge computing platforms. Each attack scenario was meticulously designed to manipulate sensor data integrity, disrupt network communications, or exhaust device resources, accurately simulating real-world IoT security threats. The experimental architecture features multiple ESP32 sensor nodes transmitting data via the UDP protocol to a centralized receiver node connected to a dedicated MATLAB-based monitoring application hosted on a PC for real-time detection analysis. The proposed IDS employs an artificial intelligence (AI) framework utilizing customized machine learning algorithms for sophisticated attack event classification and real-time threat detection. Experimental results demonstrate the superior effectiveness of the proposed IDS architecture in securing IoT and Edge device ecosystems, significantly outperforming existing detection mechanisms.

Keywords—Hardware Intrinsic Attack, IoT Security, Hardware Trojans, Artificial Intelligence, Current Profiling.

I. INTRODUCTION

This paper summarizes an innovative idea for detecting malware activity in IoT sensor networks. These networks are

typically defined by devices that are usually third-party devices and are designed to have a minimal burden of functionality and as such, not too much concern is found in terms of their security. In fact, many people will not even consider an IoT device such as an IoT camera or IoT temperature sensor, or IoT based motion sensor, or any such devices that you can find available commercially off the shelf (COTS) type of devices and therefore they are considered to be harmless in the system. However, with the advent of threats in hardware supply chains, an increased level of concern has risen related to hardware or physical layer level security in modern cyber-systems. The introduction of modern IoT based cyber-systems and their spread of use among the large population requires more research on the hardware aspect of security. It is estimated that by the end of 2025 there will be more than 21 billion IoT end devices [1]. At the same time the IoT security market is expected to grow at a compound annual growth rate of 27% an increase from \$8.6 billion in 2019 to \$58.4 billion by the end of 2027 [2], with IoT node device security alone is expected to reach a market of up to \$18.6 billion by 2027 [3]. IoT node device security has become a major concern due to the absence of concrete measures of security in IoT systems especially against the threats of HIA [4]-[7]. The COTS devices are used in most of the IoT system. AI of Things (AIoT) has been introduced recently, which combines AI and IoT with the help of a technology enabler – Edge Intelligence (EI). Since EI is perceived as a secure and privacy preserving alternative to cloud computing [8]-[12], it is a subject of growing interest to cybersecurity researchers and professionals. However, non-conventional attacks, such as HIAs, remain a major threat to EI-enabled AIoT as well. In the presented work here, we present some recent findings from an aspect of HIAs, where the IoT node, whether it's made up of a simple microcontroller type system or an FPGA based system, or it's a bit more advanced edge device, the device can be maliciously programmed either at the manufacturing level or by a third-party vendor with malicious intent. Such a device when procured will appear to be a simple normal device on unboxing and first commissioning. So, it works nicely until that hidden clock ticks to the moment where it was programmed to attack. And typically, those attacks are not noticeable to the normal user, however they can do a lot of harm in the connected system because ultimately IoT devices are sending data to your

network. This groundbreaking project aims to detect HIA in embedded systems by utilizing an innovative technique known as current profiling [13]-[15]. The motivation behind this project is the growing need to secure IoT devices, which are increasingly vulnerable to attacks that can compromise their functionality and data integrity. We know that any of these devices, whenever they are processing a program, a stored logic operation or a stored software, whether it's a microcontroller or an edge device, it consumes current, it consumes electrical energy to make its processing possible. By monitoring and analyzing this power consumption, we can distinguish between normal and abnormal device behaviors, potentially detecting malicious activity. Our proposed system builds upon the "gold standard" for device operation, which represents normal, clean operation without any malicious intent or attacks. A sensitive current monitoring sensor is connected to the device's power supply, continuously recording current consumption data over time. By comparing the collected current profile dataset of normal operation to that of abnormal processing, the system can identify signs of attack via a deployed AI machine learning-based model in a user-friendly application that was developed using MATLAB. This approach has been tested and developed in a test bed, where three different test scenarios were implemented. The first scenario involves IoT sensors powered by COTS microcontrollers, such as the ESP32. Each ESP32 is connected to a specific sensor, like temperature, humidity, soil moisture, or motion sensors. These sensors are all powered by a common power grid, with a current sensor placed in series to convert current values into voltages, which are then read by a receiver microcontroller. This data is transmitted wirelessly using UDP communication protocol to a base computer for further processing. For further system validation, a second and a third scenarios were introduced, in which the proposed detection system was tested on two edge devices including Nexys2 FPGA and Raspberry Pi 5 for detecting different intrinsic attacks, highlighting the effectiveness of the proposed IDS. In addition to this, a complete hardware setup has been developed, including an advisory power box that can be used with an IoT device powertrain to indicate malicious operations in the IoT nodes. This performance advantage stems from the innovative hardware-based detection methodology that continuously monitors the devices current consumption patterns under both normal operational and attack scenarios, establishing power consumption signatures as reliable indicators of malicious activity. The research underscores the critical importance of adaptive security mechanisms in modern IoT networks and provides comprehensive real-time monitoring capabilities with automatic threat classification, offering practical deployment solutions for cybersecurity professionals.

II. PROPOSED SOLUTION

The proposed solution was implemented on IoT and Edge devices, where four distinct cyber-attacks were executed to evaluate the effectiveness of an advanced IDS. Specifically, these attacks were launched on four IoT devices, while one attack was conducted on an Edge device. The four attack types

tested in this study include CCA, PDA, DoS, and MIMA. The IoT devices used in the experiment consisted of five ESP32 NodeMCU microcontrollers, of which four were selected for attack simulations, while one Raspberry Pi 5 and one Nexys2 FPGA were designated as the Edge devices. Each attack was conducted separately on a designated ESP32 device, while only CCA was executed on the Raspberry Pi 5 and only PDA was implemented on the Nexys2 FPGA. Each IoT and Edge device in this system was designed to serve as a standalone application for end users. The first ESP32 was connected to a soil moisture sensor, providing farmers with real-time information about the soil's water needs. The second ESP32 was connected to a PIR motion sensor, which detects movement in the monitored area. The third ESP32 was integrated with a gas sensor, ensuring gas leak detection for safety applications. The fourth ESP32 was connected to an ultrasonic sensor, measuring the distance between objects. The fifth ESP32 and Nexys2 were paired with a DHT11 sensor to monitor humidity and temperature levels. As for the Raspberry Pi 5, it was connected to a 1980×1080 high-resolution RGB camera, functioning as a real-time surveillance system. All ESP32 devices transmitted their sensor measurements to a receiver node via the UDP protocol, ensuring efficient data transmission over a shared network. The CCA was executed on the first ESP32 connected to the soil sensor, where sensor readings were duplicated and transmitted to an unauthorized channel. The PDA targeted both the second ESP32 (PIR sensor) and Nexys2 by forcing them to draw excessive current through an externally connected load, leading to battery-depletion/power-overdraw and system instability. The DoS was executed on the third ESP32 (gas sensor), where an overwhelming influx of data caused the receiver to become unresponsive to legitimate traffic. The fourth ESP32 (ultrasonic sensor) operated as a control device, experiencing normal functionality with no attack launched on it. The MIMA was deployed on the fifth ESP32 (DHT11 sensor), where temperature and humidity readings were manipulated mid-transmission between the sender and receiver nodes. On Raspberry Pi 5, the CCA attack was implemented to mirror and transmit live camera footage to an unauthorized destination, posing a significant privacy risk. The receiver node was connected to a laptop running a MATLAB-based standalone application, designed to display received data and integrate an AI-based intrusion detection model. This system effectively identifies and classifies attacks in real-time, enhancing the security of IoT and Edge computing environments. Figures 1 to 3 show the high-level architecture of the experimental setup proposed in this work. As can be seen, these architectures provide research-grade testbeds for further experiments and variations.

III. LAUNCHED ATTACKS

To evaluate the effectiveness of our proposed IDS, four attacks were implemented in this work: CCA, PDA, DoS, and MIMA. All attacks are explained next and are conceptually shown in Figure 4.

normal, 20 min CCA & PDA, 20 min Normal, 20 min CCA & DoS, 20 min Normal, 20 min CCA & MIMA, 20 min Normal, 20 min PDA & DoS, 20 min Normal, 20 min PDA & MIMA, 20 min Normal, 20 min DoS & MIMA, 20 min Normal, 20 min CCA & PDA & DoS, 20 min Normal, 20 min CCA & PDA & MIMA, 20 min Normal, 20 min CCA & DoS & MIMA, 20 min Normal, 20 min PDA & DoS & MIMA, 20 min Normal, 20 min All Attacks. So, in total we have 16 different classes including Normal class in which we are targeting to implement an AI classification model to be able to detect each one of these classes. The sequence of appearance of each attack was kept under controlled randomized order through simple random round-robin selectivity. The sampling frequency was consistently kept at 1 Hz. Next, we have split the dataset into 60% for training and 40% testing, keeping all classes balanced in numbers. From single to multiple simultaneous attacks, an overall combination grid, as shown in Figure 5, resulted in 15 attack combinations plus one Normal operation class.

CCA	PDA	DoS	MIM	CCA & PDA
CCA & DoS	CCA & MIM	PDA & DoS	PDA & MIM	DoS & MIM
CCA & PDA & DoS	CCA & PDA & MIM	CCA & DoS & MIM	PDA & DoS & MIM	All Attacks

Fig. 5.

Figure 5 displays various types of attacks combinations/classes, in addition to the Normal operation class, used in building the AI model

A window of this data showing all 16 classes is shown in Figures 6. A similar view of the CCA scenario is shown in the form of actual video vs. hacked video as seen in two monitors in Figure 7.

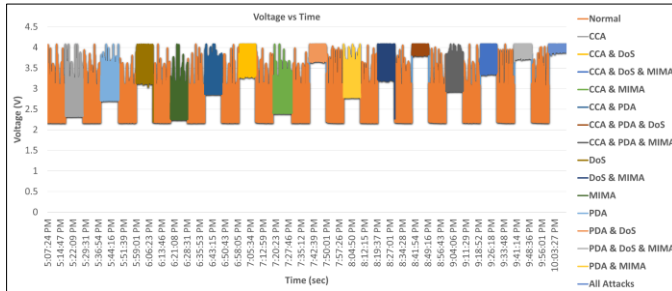


Fig. 6.

Figure 6 illustrates a waveform showing the current drawn by the IoT devices after conversion to voltage for all attacks' classes in time domain (Voltage vs Time).

The machine learning framework employs as diverse ensemble of sophisticated classifiers, including k-Nearest Neighbors (k-NN) for proximity-based pattern recognition, Ensemble Learning methods utilizing both Bagging and LPBoost algorithms for improved generalization and reduced overfitting, Support Vector Machines (SVM) for optimal hyperplane-based classification, Decision Trees (DT) for interpretable rule-based detection, and Linear Discriminant Analysis (LDA) for dimensionality reduction and feature optimization. These AI-driven classifiers are specifically

deployed on IoT sensor nodes (ESP32 microcontrollers) and edge computing platforms (Raspberry Pi 5) to leverage their computational capabilities for real-time threat analysis. Complementing this approach, the system implements a lightweight threshold-based detection mechanism on Nexys2 FPGA board. This dual-methodology framework enables comprehensive performance evaluation and comparative analysis to determine the optimal balance between detection accuracy, computational efficiency, and resource utilization, thereby providing insights into the most cost-effective and technologically sophisticated solution for diverse deployment scenarios.

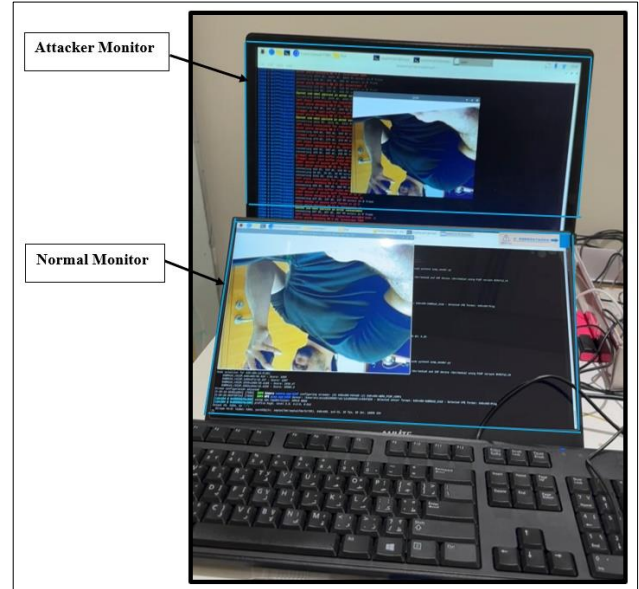


Fig. 7.

Figure 7 displays the launched CCA from Normal Raspberry Pi 5 connected monitor and Attacker's Raspberry Pi 5 monitor.

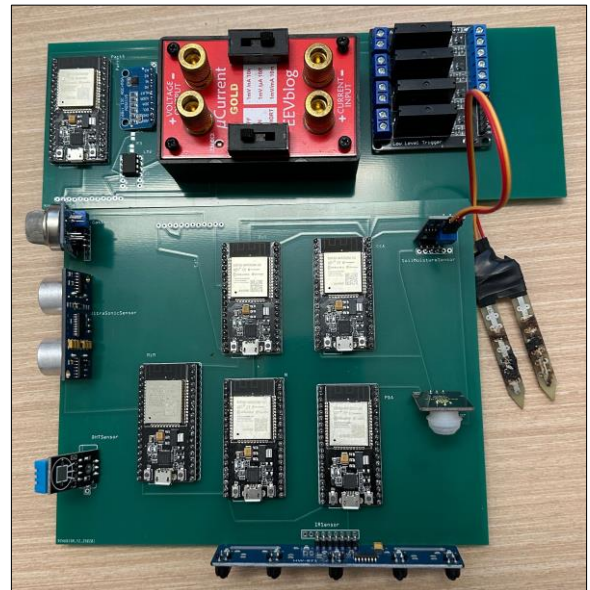


Fig. 8.

Figure 8 shows the IoT tested on a customized printed circuit board (PCB)

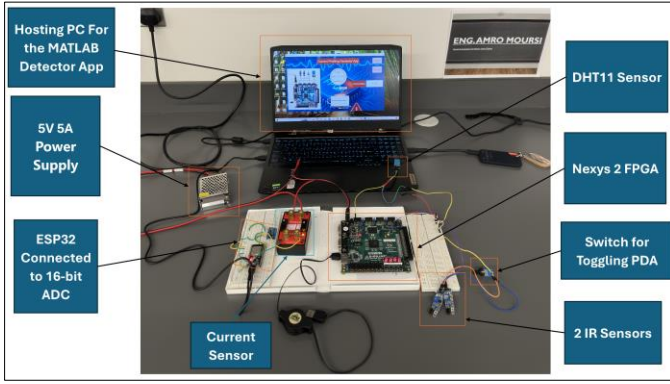


Fig. 9.

Figure 9 shows Edge devices (Nexys2 FPGA) testbed.

For classification, we have utilized Bagging (Bootstrap Aggregating), Ensemble Learning with the bagging method. The Bootstrap Aggregating (Bagging) technique is based on training multiple weak learners (typically DT), where each weak learner is trained on a randomly drawn subset (with replacement) of the original dataset. This ensures diversity among the learners, which helps in reducing variance and improving generalization. Once each weak learner is trained, the final prediction is obtained through majority voting in the classification task. The key advantages of this approach include reducing overfitting, as combining multiple weak learners helps to smooth out individual model biases, and parallelism, since each weak learner is trained independently, making bagging computationally efficient when implemented in parallel. This makes ensemble methods like bagging effective for improving performance, stability, and robustness in machine learning models. Ablation experiments were conducted utilizing different hyper parameters for further system verification for optimal results across different machine learning classifiers, available in Tables I-III. The experiments show the k-NN model achieved its best performance with k=5 neighbors using City Block distance, reaching 90.15% accuracy. In contrast, the SVM models performed poorly, with both Gaussian and RBF kernels yielding only 66.74% accuracy. The Fit Ensemble (Bagging) method demonstrated outstanding results for 300 learning cycles or higher, achieving 99.78% accuracy. Table IV shows the summarized results of the top models' accuracies. Comparison factors used to evaluate the proposed machine learning classifiers were calculated utilizing equations 1-4.

$$Accuracy = \frac{\sum_{i=1}^n C_{i,i}}{\sum_{i=1}^n \sum_{j=1}^n C_{i,j}} \quad (1)$$

$$Mean Precision = \frac{1}{n} \sum_{i=1}^n \frac{C_{i,i}}{\sum_{j=1}^n C_{j,i}} \quad (2)$$

$$Mean Recall = \frac{1}{n} \sum_{i=1}^n \frac{C_{i,i}}{\sum_{j=1}^n C_{i,j}} \quad (3)$$

$$Mean F1_Score = \frac{1}{n} \sum_{i=1}^n 2 \times \frac{\frac{C_{i,i}}{\sum_{j=1}^n C_{j,i}} \times \frac{C_{i,i}}{\sum_{j=1}^n C_{i,j}}}{\frac{C_{i,i}}{\sum_{j=1}^n C_{j,i}} + \frac{C_{i,i}}{\sum_{j=1}^n C_{i,j}}} \quad (4)$$

TABLE I. FIT ENSEMBLE (BAGGING) ABLATION EXPERIMENT

# Learning Cycles	Accuracy	Precision	Recall	F1_Score
100	99.12%	98.51%	98.33%	98.32%
300	99.78%	99.61%	99.58%	99.58%
500	99.78%	99.61%	99.58%	99.58%
700	99.78%	99.61%	99.58%	99.58%
900	99.78%	99.61%	99.58%	99.58%

TABLE II. KNN ABLATION EXPERIMENT

#Neighbors	Distance weight	Distance	Accuracy	Precision	Recall	F1_Score
5	Equal	Euclidean	71.55%	NaN	47.71%	NaN
	inverse	Euclidean	72.43%	NaN	49.40%	NaN
	Squared Inverse	Euclidean	72.65%	NaN	49.82%	NaN
	Equal	City Block	90.15%	88.03%	81.65%	81.84%
	Equal	Jaccard	86.21%	78.20%	73.75%	72.92%
	Equal	Hamming	86.21%	78.20%	73.75%	72.92%

TABLE III. SVM ABLATION EXPERIMENT

Kernel	Accuracy	Precision	Recall	F1_Score
Gaussian	66.74%	NaN	38.33%	NaN
Linear	57.55%	18.68%	21.56%	NaN
RBF	66.74%	NaN	38.33%	NaN
Polynomial	51.64%	NaN	12.83%	NaN

TABLE IV. PERFORMANCE COMPARISON OF THE PROPOSED MACHINE LEARNING CLASSIFIERS.

Classifier	Accuracy	Precision	Recall	F1_Score
Fit ensemble (Bagging)	99.78%	99.61%	99.58%	99.58%
Fit ensemble (LPB)	98.03%	96.62%	96.40%	96.36%
k-NN	90.15%	88.03%	81.65%	81.84%
LDA	84.46%	75.17%	74.56%	74.50%
DT	82.71%	70.66%	68.70%	67.78%
SVM	66.74%	NaN	38.33%	NaN

V. USER INTERFACE

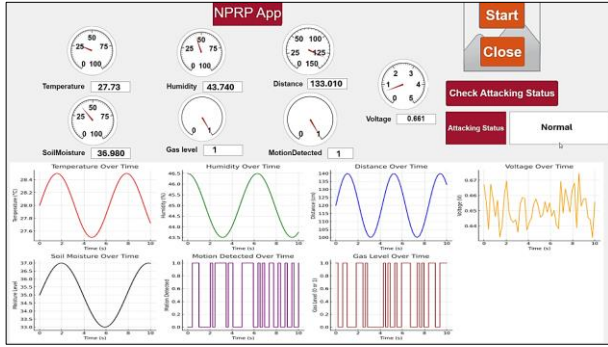


Fig. 10.

Figure 10 shows the proposed IoT-MATLAB application displaying the sensors readings alongside with the current consumption after conversion to voltage, as well as the current system status whether Normal or Malicious state.

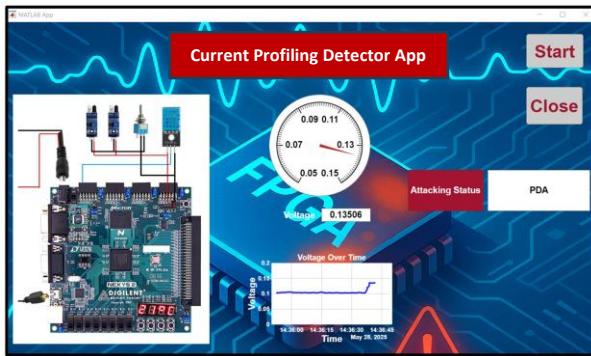


Fig. 11.

Figure 11 shows the proposed Edge devices (Nexys2)-MATLAB application displaying the current consumption after conversion to voltage, as well as the current system status whether Normal or Malicious state.

VI. CONCLUSION

In conclusion, this study successfully demonstrated the effectiveness of an AI-driven IDS for addressing the gap in detecting intrinsic cyber threats in IoT and Edge computing environments. By employing Ensemble Learning (Bagging and LPBoost), the system effectively classified cyberattacks such as CCA, PDA, DoS, and MIMA. The classification results indicate that ensemble learning techniques (Bagging and LPBoost) provided the highest accuracy, with Bagging achieving 99.78% accuracy and LPBoost reaching 98.03% accuracy, demonstrating their robustness in real-time threat detection. k-NN, LDA, and DT exhibited moderate classification performance, with 90.15%, 84.46%, and 82.71% accuracy, respectively, while SVM performed the worst, with accuracy levels of 66.74%. These findings highlight the significance of ensemble learning methods in securing IoT ecosystems against cyber threats in addition to the user-friendly application interface that provides easy access for the users to easily detect whether the devices are in malicious status. Future research work will focus on enhancing the adaptability of the IDS to include mitigation techniques and further optimizing the machine learning models to improve the detection accuracy of cyberattacks in IoT and Edge devices environments.

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